

Wear Your Future

"During a Formula 1 race, a driver experiences wrenching forces of more than 4.5G. His heart rate may exceed 180 beats per minute and his blood pressure could rise by half. With soaring temperatures inside the cramped cockpit he will also dehydrate, typically losing 2-3 litres of water during the race. Yet the driver must concentrate well enough to achieve lap times that might vary by just a tenth of a second. This is tough, on both mind and body. Hence it is not just the performance of the car itself which an array of sensors keeps an eye on, wirelessly transmitting data about the engine, suspension and so on to the pit crews. The drivers' own vital signs are constantly monitored, too." —Economist

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What are wearable computers all about

Way back in 1961 two mathematicians, Edward O. Thorp and Claude Shannon, developed miniature computerised timing devices to help them cheat at the game of roulette. Later, *The Eudaemonic Pie* - a book written by American author Thomas A. Bass in 1985 tells the story about a few University of California, physics graduate students. They develop very-small-size computers which are clandestinely worn on modified platform soled shoes, to predict casino roulette games.

In the early 1980s, Steve Mann, one of the forerunners of wearable technology, produced a backpack-mounted computer to control photographic equipment. Later in 1994 he developed a headset that was able to send images to the Internet. A few years later IBM carried out some trials on wearable computing on the ThinkPad, and by 2001 it presented the first model of a wristwatch computer known then as the WatchPad. Thereafter several interesting developments took place in the field of wearable computing.

Wearable computers, or body-borne computers, are small electronic devices which are worn by the user under, with or on top of clothing. This type of wearable technology is especially useful for applications that demand advanced computational support, that is, more than hardware-coded logic. Since there is continuous communication between the



IBM Linux Watch (IBM Linux Watch, 2000) & IBM watch pad 1.5 (IBM WatchPad 1.5, 2001) (Source: <http://www.itworld.co.kr/>)



Anaesthesiologist keeps his attention on the patient while viewing vital signs via Google Glass (Source: <http://www.medical.philips.com/main/about/future-of-healthcare/>)



A US Army Land Warrior soldier (Source: http://www.army-technology.com/projects/land_warrior/)



Wearable electronic sensors on a finger (Source: <http://jasminepark182.com/>)

wearable device and user, it is 'always on' so that there is no botheration of switching off. Wearable-computer-based devices seamlessly merge with our daily lives in the sense that we don't need to stop doing something to operate the device. Rather the wearable devices act more as an extension of our mind and body and sometimes also in a way like an artificial body part.

War with wearable technology

Since the mid-1990s, the US military has been refining prototype soldier

systems under its Land Warrior programme. A computer manages a day-light video electro-optic sight, a thermal sight mounted on the soldier's M-4 carbine, a multichannel wireless communications system, a helmet-mounted display system and lightweight body armour. From the helmet-mounted display, a soldier can view his environment through the sensors mounted on his rifle. Using GUI, he can switch to a zoom-able map, where he can plot his own position, determine the locations of his squad members and locate suspected and confirmed enemy positions.